

PHYSICS

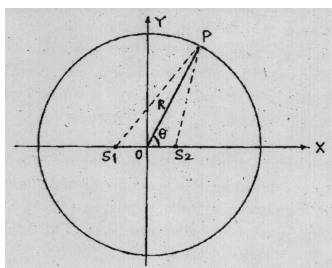
TARGET : JEE- Advanced

CAPS-22

WAVE OPTICS

SCQ (Single Correct Type) :

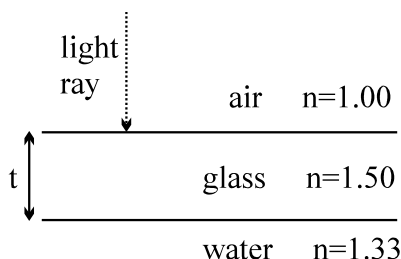
- In a Young's double slit experiment, the separation between the slits is d , distance between the slit and screen is D ($D \gg d$). In the interference pattern, there is a maxima exactly in front for each slit. Then the possible wavelength(s) used in the experiment are
 (A) $d^2/D, d^2/2D, d^2/3D$ (B) $d^2/D, d^2/3D, d^2/5D$
 (C) $d^2/2D, d^2/4D, d^2/6D$ (D) none of these
- Two coherent sources of light S_1 and S_2 , equidistant from the origin, are separated by a distance 2λ as shown. They emit light of wavelength λ . Interference is observed on a screen placed along the circle of large radius R . Point is seen to be a point of constructive interference. Then angle θ (other than 0° and 90°) is



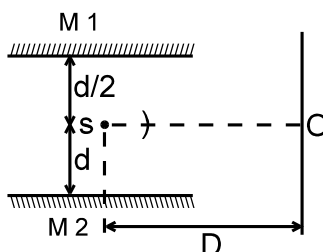
- (A) 45° (B) 30°
 (C) 60° (D) Not possible in the first quadrant
- In a regular YDSE, when thin film of refractive index μ is placed in front of the upper slit then it is observed that the intensity at the central point becomes half of the original intensity. It is also observed that the initial 3rd maxima is now below the central point and the initial 4th minima is above the central point. Now, a film of refractive index μ_1 and thickness same as the above film, is put in the front of the lower slit also. It is observed that whole fringe pattern shifts by one fringe width. What is the value of μ_1 ?
 (A) $(4\mu+9)/12$ (B) $(4\mu+9)/13$ (C) $(4\mu+9)/11$ (D) None
 - Large number (N) of coherent waves superimpose at a point in the medium represented by-
 $y_1 = a \sin(kx - \omega t)$
 $y_2 = a \sin(kx - \omega t + \phi)$
 $y_3 = a \sin(kx - \omega t + 2\phi)$

 $y_N = a \sin(kx - \omega t + N\phi)$
 where ϕ is very small. Then amplitude of resultant wave is-
 (A) zero (B) Na (C) $(a/\phi) \sin(N\phi/2)$ (D) $2(a/\phi) \sin(N\phi/2)$

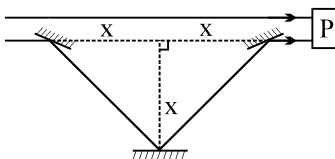
5. Many people's glasses appear to be a blue-green colour when viewed under reflected light. A thin film of index of refraction $n = 1.35$ is applied to the outside surface of the glass so that the film/glass interface does not reflect any red light incident near normal of wavelength $\lambda = 630$ nm. What thickness must the film layer be in order to achieve this? Take the index of refractions of air and glass to be 1.0 and 1.6 respectively.
 (A) 157.5 nm (B) 315.0 nm (C) 233.3 nm (D) 116.7 nm
6. A light ray is incident normal to a thin layer of glass. Given the figure, what is the minimum thickness of the glass that gives the reflected light an orangish color ($\lambda_{\text{air}} = 600$ nm)?



- (A) 50 nm (B) 100 nm (C) 150 nm (D) 200 nm
7. M_1 and M_2 are plane mirrors and kept parallel to each other. At point O there will be a maxima for wavelength. Light from monochromatic source S of wavelength λ is not reaching directly on the screen. Then λ is : [$D \gg d$, $d \gg \lambda$]



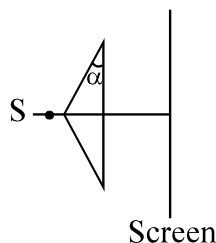
- (A) $\frac{3d^2}{D}$ (B) $\frac{3d^2}{2D}$ (C) $\frac{d^2}{D}$ (D) $\frac{2d^2}{D}$
8. Microwaves of wavelength $\lambda = 5$ cm and intensity I_0 are split and recombined by the metallic mirror system as shown in the figure. The intensity of the microwave at the detector P is minimum if x is



- (A) 0.88 cm (B) 3.54 cm (C) 3.02 cm (D) 6.04 cm
9. An YDSE is performed with bi-chromatic light ($5500\text{\AA}$ and $6000\text{\AA}$) for $d = 2$ mm and $D = 1$ m. Distance of first complete maxima from the central maxima on the screen, is
 (A) 1.1 mm (B) 2.2 mm (C) 3.3 mm (D) 4.4 mm

MCQ (One or more than one correct) :

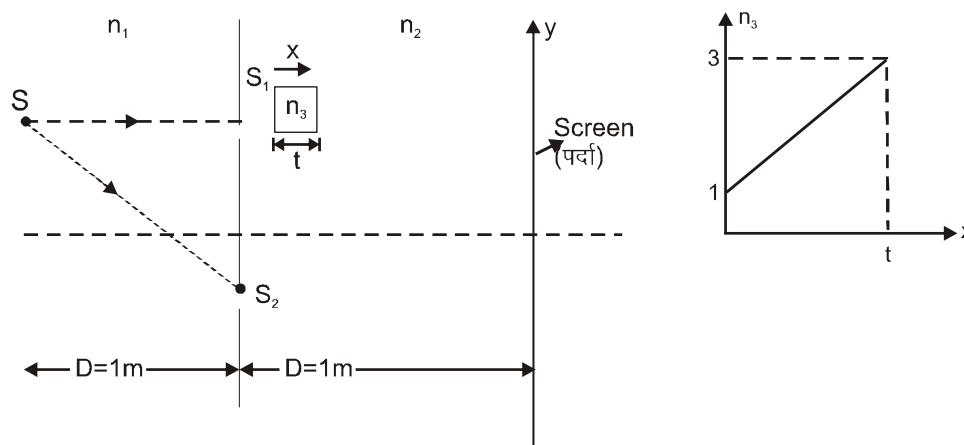
10. For the biprism experiment shown in the figure, the fringe width increases when



- (A) biprism is moved towards the slit
 (B) entire apparatus is submerged in a liquid having R.I. less than that of prism
 (C) a biprism having smaller angle α is used
 (D) the slit width is reduced

Comprehension Type Question:

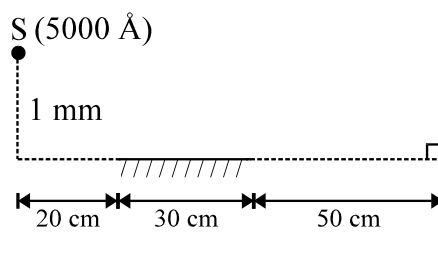
In YDSE arrangement as shown in figure, fringes are seen on screen using monochromatic source S having wavelength $3000 \text{ \AA}$ (in air). S_1 and S_2 are two slits separated by $d = 1 \text{ mm}$ and $D = 1 \text{ m}$. Left of slits S_1 and S_2 medium of refractive index $n_1 = 2$ is present and to the right of S_1 and S_2 medium of $n_2 = 3/2$ is present. A thin slab of thickness 't' is placed in front of S_1 . The refractive index of n_3 of the slab varies with distance from its starting face as shown in figure.



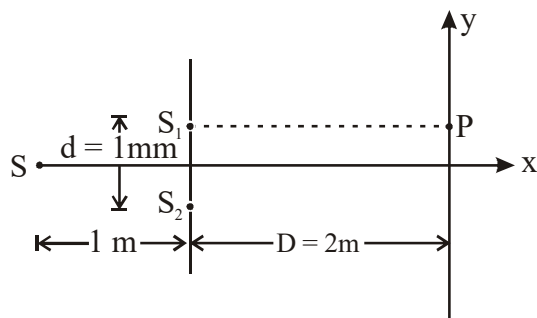
11. In order to get central maxima at the centre of screen, the thickness of slab required is :
 (A) $1 \text{ }\mu\text{m}$ (B) $2 \text{ }\mu\text{m}$ (C) $0.5 \text{ }\mu\text{m}$ (D) $1.5 \text{ }\mu\text{m}$
12. If thickness of the slab is selected $1 \text{ }\mu\text{m}$, then position of central maxima will be: (y-coordinate)
 (A) $\frac{1}{3} \text{ mm}$ (B) $-\frac{1}{3} \text{ mm}$ (C) $\frac{1}{6} \text{ mm}$ (D) $-\frac{1}{6} \text{ mm}$
13. Fringe width on the screen is :
 (A) 0.4 mm (B) 0.1 mm (C) 0.2 mm (D) 0.3 mm

Numerical based Questions :

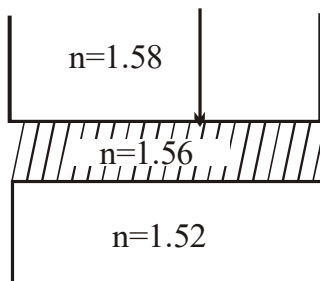
14. To reduce the light reflected by the glass surface of a camera lens, the surface is coated with a thin layer of another material which has an index of refraction ($\mu = 7/4$) smaller than that of glass. The least thickness of the layer, to ensure that light falling perpendicularly on the surface and having wavelengths, $\lambda_1 = 700\text{nm}$ and $\lambda_2 = 420\text{nm}$ will be weakly reflected for both wavelengths is $x \times 10^{-8}\text{m}$. Find x ?
15. Find the total number of fringes formed on screen in the Lloyd's mirror arrangement shown.



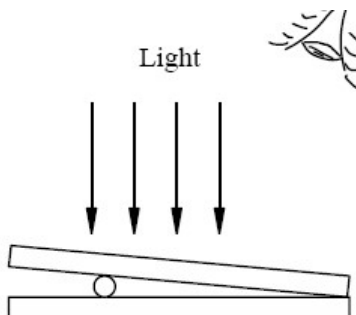
16. In a modified YDSE the source S of wavelength $5000\text{\AA}$ oscillates about axis of setup according to the equation $y = 0.5 \sin\left(\frac{\pi}{6}\right)t$, where y is in millimeter and t in second. At what time t will the intensity at P , a point exactly in front of slit S_1 , be maximum for the first time ?



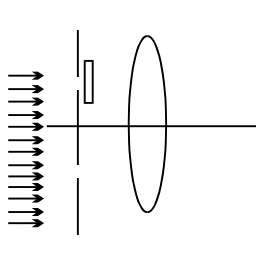
17. A thin film of plastic ($n = 1.56$) is $0.25\text{ }\mu\text{m}$ thick. It is sandwiched between two glass slabs with refractive indices of 1.58 and 1.52, respectively. White light ($400 - 700\text{ nm}$) is first incident normally on the slab for which $n = 1.58$. Which visible wavelength (in nm) is missing in the reflected light ?



18. A very fine hair-like fiber is stuck between two microscope slides. As a result of this, there is a wedge of air between them. When the slides are illuminated normally (from above) with light of wavelength 550 nm, bright and dark interference bands are formed. The fiber is seen to lie at the position of fifth dark band counting from the common edge. What is the diameter (in nm) of the fiber?

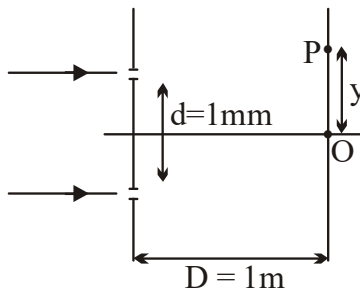


19. The intensity received at the focus of the lens is I when no glass slab has been placed in front of the slit. Both the slits are of the same dimension and the plane wavefront incident perpendicularly on them, has wavelength λ . On placing the glass slab, the intensity reduces to $3I/4$ at the focus. Find out the minimum thickness of the glass slab (in Å) if its refractive index is $3/2$. Given $\lambda = 6933 \text{ Å}$, $\mu = 1.5$.



Matrix Match Type :

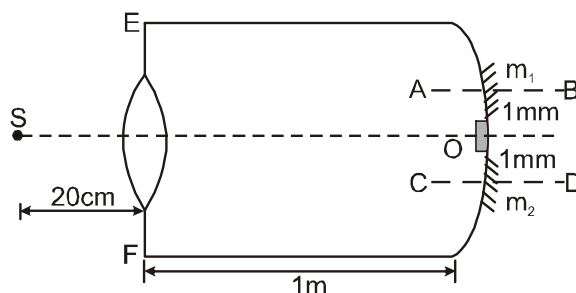
20. A parallel beam of visible light consisting of wavelengths λ_1 & λ_2 is incident on a standard YDSE apparatus with $d = 1 \text{ mm}$, $D = 1 \text{ m}$. P is a point on the screen at a distance y from center of screen O. $y = y_1$ is the nearest point above O where the two maxima coincide. $y = y_2$ is the nearest point above O where the two minima coincide. β_1 & β_2 are fringe width corresponding to wave length λ_1 & λ_2 .



Column-I	Column-II
(A) $\beta_1 = 0.3 \text{ mm}, \beta_2 = 0.5 \text{ mm}$	(P) The 2 nd nearest point above O where two maxima coincide is $y = 2y_1$
(B) $\beta_1 = 0.3 \text{ mm}, \beta_2 = 0.4 \text{ mm}$	(Q) y_2 has no finite value
(C) $\beta_1 = 0.2 \text{ mm}, \beta_2 = 0.4 \text{ mm}$	(R) The 2 nd nearest point above O where the two minima coincide is $y = 3y_2$.
(D) $\beta_1 = 0.2 \text{ mm}, \beta_2 = 0.6 \text{ mm}$	(S) $y_1 = \text{LCM of } \beta_1 \text{ \& } \beta_2$
	(T) The 2 nd nearest point above O where two minima coincide is $y = 2y_2$.

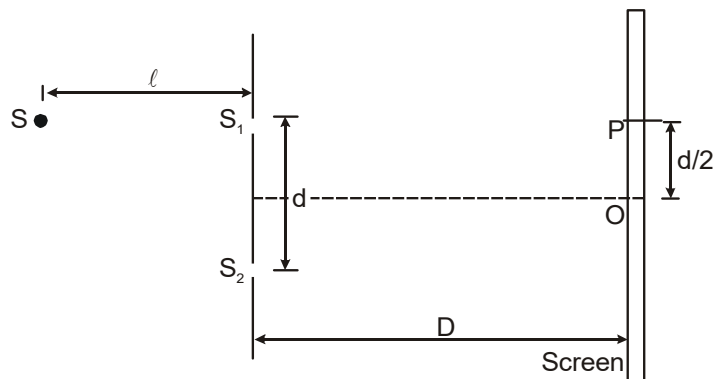
Subjective Type Questions :

21. An equi convex lens of focal length 10 cm (in air) and R.I. $3/2$ is put at a small opening on a tube of length 1 m fully filled with liquid of R.I. $4/3$. A concave mirror of radius of curvature 20 cm is cut into two halves m_1 and m_2 and placed at the end of the tube. m_1 & m_2 are placed such that their principal axis AB and CD respectively are separated by 1 mm each from the principal axis of the lens. A slit S placed in air illuminates the lens with light of frequency $7.5 \times 10^{14} \text{ Hz}$. The light reflected from m_1 and m_2 forms interference pattern on the left end EF of the tube. O is an opaque substance to cover the hole left by m_1 & m_2 . Find :

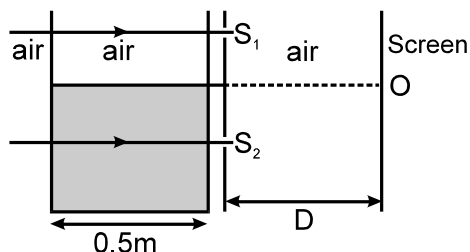


- (a) the position of the image formed by lens water combination.
 (b) the distance between the images formed by m_1 & m_2 .
 (c) width of the fringes on EF.
22. In a YDSE experiment, the distance between the slits & the screen is 100 cm. For a certain distance between the slits, an interference pattern is observed on the screen with the fringe width 0.25 mm. When the distance between the slits is increased by $\Delta d = 1.2 \text{ mm}$, the fringe width decreased to $n = 2/3$ of the original value. In the final position, a thin glass plate of refractive index 1.5 is kept in front of one of the slits & the shift of central maximum is observed to be 20 fringe width. Find the thickness of the plate & wavelength of the incident light.

23. A source S is kept directly behind the slit S_1 in a double-slit apparatus. Find the phase difference at a point O which is equidistant from S_1 & S_2 . What will be the phase difference at P if a liquid of refractive index μ is filled; (wavelength of light in air is λ due to the source). Assume same intensity due to S_1 and S_2 on screen and position at liquid.
- ($\lambda \ll d, d \ll D, \ell \gg d$)



- (a) between the screen and the slits.
- (b) between the slits & the source S . In this case find the minimum distance between the points on the screen where the intensity is half the maximum intensity on the screen.
24. A monochromatic light of $\lambda = 5000 \text{ \AA}$ is incident on two slits separated by a distance of $5 \times 10^{-4} \text{ m}$. The interference pattern is seen on a screen placed at a distance of 1 m from the slits. A thin glass plate of thickness $1.5 \times 10^{-6} \text{ m}$ & refractive index $\mu = 1.5$ is placed between one of the slits & the screen. Find the intensity at the centre of the screen, if the intensity there is I_0 in the absence of the plate. Also find the lateral shift of the central maximum.
25. In a YDSE, radiowaves of wavelength $\lambda = 50 \text{ mm}$ (in air) are used. Two parallel beams are first passed through a glass vessel of base length 0.5 m as shown, filled upto some height (shaded region) with a liquid whose refractive index is varying with distance ' x ' (in meters) from the left face as $\mu(x) = \mu_0(a + x)$ where $\mu_0 = 4/3 \text{ m}^{-1}$ and $a = 1 \text{ m}$. If intensity due to each slit at O (symmetrical with respect to S_1 and S_2) is I . Find the intensity at point O on the screen.



ANSWER KEY (CAPS-22)

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|---|------------|----------|----------|-----------|
| 1. (C) | 2. (C) | 3. (B) | 4. (D) | 5. (D) |
| 6. (B) | 7. (B) | 8. (D) | 9. (C) | 10. (ABC) |
| 11. (B) | 12. (B) | 13. (C) | 14. 30 | 15. 12 |
| 16. 1 | 17. 520 nm | 18. 1100 | 19. 2311 | |
| 20. (A) P, R, S (B) P, Q, S (C) P, Q, S (D) P, R, S | | | | |
| 21. (a) 80 cm behind the lens (b) 4 mm (c) $\beta = 60 \mu\text{m}$ 22. $\lambda = 600 \text{ nm}$, $t = 24 \mu\text{m}$ | | | | |
| 23. (a) $\Delta\phi = \left(\frac{1}{\ell} + \frac{\mu}{D}\right) \frac{\pi d^2}{\lambda}$ (b) $\Delta\phi = \left(\frac{\mu}{\ell} + \frac{1}{D}\right) \frac{\pi d^2}{\lambda}$; $D_{\min} = \frac{\beta}{2} = \frac{\lambda D}{2d}$ 24. 0, 1.5 mm 25. I | | | | |